

AISI 1020 Steel, as rolled, 25 mm (1 in.) round**Categories:** [Metal](#); [Ferrous Metal](#); [Carbon Steel](#); [AISI 1000 Series Steel](#); [Low Carbon Steel](#)**Material Notes:** Applications include parts in the case hardened condition where core strength is not critical, and for shafts of larger cross section that are not highly stressed. Other uses include lightly stressed gears with hard wearing surfaces and case hardened pins and chains.**Key Words:** UNS G10200, AMS 5032, AMS 5045, ASTM A29, ASTM A108, ASTM A510, ASTM A519, ASTM A29, A108, A510, A512, A513, A519, A544, A575 M10120, A576, A635, A659, A827, A830, BS 970 040A20, 050A20 (En2C), 050A20 (En2D), 060A20, BS 970 Part 1 070M20, DEF STAN95-1-1 C1020, SAE J412, SAE J414, DIN 1.0402, AFNOR CC 20, UNI C 20, SS14 1450 (Sweden), SAE J403**Vendors:** No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.87 g/cc	0.284 lb/in ³	Typical for steel
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	137	137	
Hardness, Knoop	156	156	Converted from Brinell
Hardness, Rockwell B	75	75	Converted from Brinell
Hardness, Vickers	143	143	Converted from Brinell
Tensile Strength, Ultimate	472 MPa	68500 psi	
Tensile Strength, Yield	384 MPa	55700 psi	
Elongation at Break	32 %	32 %	in 50 mm
Reduction of Area	66.5 %	66.5 %	
Modulus of Elasticity	186 GPa	27000 ksi	
Bulk Modulus	148 GPa	21500 ksi	calculated from elastic modulus and Poisson's ratio
Poissons Ratio	0.29	0.29	
Shear Modulus	72.0 GPa	10400 ksi	calculated from elastic modulus and Poisson's ratio
Izod Impact	85.0 J	62.7 ft-lb	
Charpy Impact 	16.9 J	12.5 ft-lb	
	@Temperature -30.0 °C	@Temperature -22.0 °F	
	18.0 J	13.3 ft-lb	
	@Temperature -18.0 °C	@Temperature -0.400 °F	
	20.0 J	14.8 ft-lb	
	@Temperature -3.00 °C	@Temperature 26.6 °F	
	24.0 J	17.7 ft-lb	
	@Temperature 10.0 °C	@Temperature 50.0 °F	
	41.0 J	30.2 ft-lb	
	@Temperature 38.0 °C	@Temperature 100 °F	
	54.0 J	39.8 ft-lb	
	@Temperature 65.0 °C	@Temperature 149 °F	
	61.0 J	45.0 ft-lb	
	@Temperature 95.0 °C	@Temperature 203 °F	
	68.0 J	50.2 ft-lb	
	@Temperature 150 °C	@Temperature 302 °F	

Electrical Properties	Metric	English	Comments
Electrical Resistivity 	0.0000159 ohm-cm @Temperature 0.000 °C	0.0000159 ohm-cm @Temperature 32.0 °F	condition unknown
	0.0000219 ohm-cm @Temperature 100 °C	0.0000219 ohm-cm @Temperature 212 °F	condition unknown
	0.0000292 ohm-cm @Temperature 200 °C	0.0000292 ohm-cm @Temperature 392 °F	condition unknown

Thermal Properties	Metric	English	Comments
CTE, linear 	11.7 µm/m-°C @Temperature 20.0 - 100 °C	6.50 µin/in-°F @Temperature 68.0 - 212 °F	
	12.1 µm/m-°C @Temperature 20.0 - 200 °C	6.72 µin/in-°F @Temperature 68.0 - 392 °F	
	12.8 µm/m-°C @Temperature 20.0 - 300 °C	7.11 µin/in-°F @Temperature 68.0 - 572 °F	
	13.3 µm/m-°C @Temperature 20.0 - 400 °C	7.39 µin/in-°F @Temperature 68.0 - 752 °F	
	13.9 µm/m-°C @Temperature 20.0 - 500 °C	7.72 µin/in-°F @Temperature 68.0 - 932 °F	
	14.4 µm/m-°C @Temperature 20.0 - 600 °C	8.00 µin/in-°F @Temperature 68.0 - 1110 °F	
	14.8 µm/m-°C @Temperature 20.0 - 700 °C	8.22 µin/in-°F @Temperature 68.0 - 1290 °F	
Specific Heat Capacity 	0.486 J/g-°C @Temperature >=100 °C	0.116 BTU/lb-°F @Temperature >=212 °F	condition unknown
	0.519 J/g-°C @Temperature 150 - 200 °C	0.124 BTU/lb-°F @Temperature 302 - 392 °F	
	0.599 J/g-°C @Temperature 350 - 400 °C	0.143 BTU/lb-°F @Temperature 662 - 752 °F	
Thermal Conductivity	51.9 W/m-K	360 BTU-in/hr-ft²-°F	Typical steel

Component Elements Properties	Metric	English	Comments
Carbon, C	0.17 - 0.23 %	0.17 - 0.23 %	
Iron, Fe	99.08 - 99.53 %	99.08 - 99.53 %	As remainder
Manganese, Mn	0.30 - 0.60 %	0.30 - 0.60 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.